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Basic Principles of Renal Replacement Therapy

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Acute kidney injury (AKI) is a common complication in critically ill patients admitted to the intensive care unit (ICU).[1–3] Epidemiologic data suggest that over 50% of ICU patients suffer from AKI, and up to 13.5% will be treated with renal replacement therapy (RRT).[1–3] Change in patient characteristics, with admission of older patients with more comorbidities such as diabetes, cardiovascular disease, and hypertension, resulted over the last decade in a marked increase of the proportion of patients treated with RRT.[4,5] RRT has therefore become an essential and often used treatment option for ICU patients, involving a spectrum of treatment modalities, with various advantages and disadvantages depending on the situation. RRT is most often delivered via extracorporeal techniques, but the COVID-19 pandemic caused renewed interest in peritoneal dialysis (PD), a technique using the peritoneum of the patient as a semipermeable exchange membrane.

TECHNICAL ASPECTS OF RRT

Extracorporeal techniques can be done with different modalities (Table 38.1). These are named according to the duration of RRT and the technique used for exchange of solutes and water: diffusion or convection.

Diffusion and Convection

Exchange of waste products over the semipermeable membrane can be via diffusion (hemodialysis) or convection (hemofiltration) (Fig. 38.1). In diffusion, blood and dialysate flow countercurrent on both sides of the semipermeable membrane of the hemofilter. The driving force that moves solutes over the semipermeable membrane is the solute concentration gradient. Uremic toxins such as blood urea nitrogen and creatinine will have high blood concentrations and are absent in the dialysate. Other factors that determine the movement of solutes from blood to dialysate are the diffusion coefficient of the membrane, its thickness, and its surface area. Diffusion is very efficient in the removal of small molecules such as potassium, ammonium, and creatinine (<500 Da); it is less efficient in the removal of larger solutes.

In hemofiltration, solutes and water are transported over the membrane by a difference in pressure between both sides of the membrane. Water and solutes are pressed from the blood compartment to the so-called effluent. The permeability coefficient of the membrane and the difference in pressure between both sides of the membrane determine the amount of fluid and solutes transported over the membrane. The effluent rate is controlled by a pump. Hemofiltration is more efficient for removal of larger molecules. In hemodiafiltration, both convection and diffusion are combined.

There are currently no data to suggest the superiority of diffusion over convection.

Duration of RRT

Intermittent hemodialysis (IHD) is a very efficient dialysis technique performed during a 3- to 4-hour period; continuous renal replacement therapy (CRRT) is less efficient, but done 24 hours a day; and hybrid techniques, alternatively termed *sustained low-efficiency daily dialysis* or *extended daily dialysis* (SLEDD or EDD), have intermediate efficacy and are used 6–12 hours per day.[6,7]

Intermittent and hybrid therapies will be performed with the use of dialysis machines that are also used for chronic dialysis patients. These monitors typically have more complicated interfaces and are therefore often managed by dialysis nurses. CRRT is most often performed with specifically designed monitors with a relatively easy interface and are managed by ICU nurses. Some centers also use these RRT machines for hybrid therapy.

Specifics Aspects of a CRRT Circuit

Fig. 38.2 shows the different aspects of CRRT performed as continuous venovenous hemofiltration (CVVH), continuous venovenous hemodialysis (CVVHD), and continuous venovenous hemodiafiltration (CVVHDF).

Dose of RRT

The dose of CRRT is by convention expressed as the clearance of urea, a small molecule that both in hemodialysis and hemofiltration is not retained by the membrane. The effluent rate—the volume of fluid produced by hemofiltration and/or hemodialysis—therefore equals the clearance of urea, and when corrected for body weight, can be used to express the dose of CRRT: dose of CRRT = effluent rate per hour per kg body weight (mL/kg/h). For a desired dose of 25 mL/kg/h for a 60-kg patient, the effluent rate will be $25 \times 60 = 1500$ mL/h. This effluent needs to be partly or completely replaced by a replacement fluid; otherwise, fluid losses will be too high. The amount of effluent that is replaced will determine the fluid balance of the patient. This replacement fluid can be given prefilter (predilution), postfilter (postdilution), or in a combination of both. When given in postdilution mode, blood will concentrate while passing through the capillaries of the hemofilter. This may lead to clogging (partial clotting) and clotting of the capillaries, leading to decreased efficacy because fewer capillaries are available. To prevent this, a filtration fraction—the proportion of effluent over plasma flow—of less than 25% is advised. The filtration fraction is indicated on the dashboard of present-day CRRT machines.

In CRRT, the delivered dose should be 20–25 mL/kg/h of effluent. Two large prospective randomized studies compared this dose to a higher dose and found that outcomes were similar.[8,9] Treatment interruptions often lead to a lower delivered dialysis dose. Therefore it is advised to prescribe a dose of 25–30 mL/kg/h.[10]

TABLE 38.1 Renal Replacement Therapy Modalities

Modality	Abbreviation	Treatment Duration (per day)	Blood Flow (mL/h)	Dialysate Flow (mL/min)
Intermittent hemodialysis	IHD	2–4 h	200–450	
Hybrid techniques • Slow low-efficiency daily dialysis • Extended daily dialysis • Prolonged intermittent renal replacement therapy	 SLEDD EDD PIRRT	6–12 h	150–200	One and a half times blood flow
Continuous renal replacement therapy • Continuous venovenous hemofiltration • Continuous venovenous hemodialysis • Continuous venovenous hemodiafiltration	CRRT CVVH CVVHD CVVHDF	24 h	100–250	
Peritoneal dialysis • Manual exchanges • Automated PD (cycler)	PD CAPD APD	Solution: dextrose 1.5% or glucose 1.36% to start, higher glucose concentration according to ultrafiltration needs Fill volume 1500–2000 mL Dwell time per exchange: 2 h, consider change to 4 h once acidosis, pulmonary edema, and hyperkalemia are resolved		

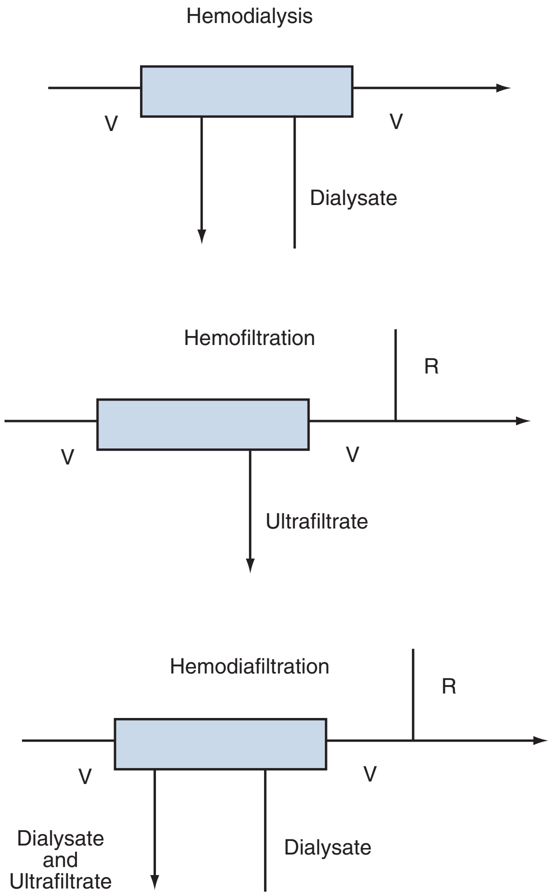

Fig. 38.1 Diffusion (as in hemodialysis) and convection (as in hemofiltration). *R*, Replacement fluid; *V*, venous blood prefilter and postfilter.

In intermittent RRT, the minimum delivered dose of dialysis should be three sessions of at least 3 hours per week, with a blood flow of >200 mL/min and a dialysate flow of >500 mL.[11]

Anticoagulation

During RRT, the patient's blood is purified by circulating it through an extracorporeal circuit containing a hemodialyzer, consisting of about 10,000 semipermeable capillary fibers. Disordered blood flow and contact of the blood with the bioincompatible material of the fibers activate the coagulation cascade, leading to clotting and subsequent blocking of the capillary fibers. This results in a reduction of the efficiency of the dialysis procedure, as fewer fibers remain available for solute exchange. When more pronounced, coagulation can even cause loss of extracorporeal blood volume by complete clotting of the extracorporeal circuit, so blood can no longer be returned to the patient.[12] Anticoagulation for RRT should be tailored according to patient characteristics and the modality chosen. Capturing the end result of the rebalanced hemostatic system in ICU patients with AKI is challenging, but new laboratory testing methods, including thromboelastography/rotational thromboelastometry (TEG/ROTEM) and thrombin generation assays, potentially yield vital information. The first technique assesses development, strength, and dissolution of clots by viscoelastic testing and does not take long before providing results. It offers information on the contribution of both enzymatic and cellular components in whole blood. Thromboelastometry analysis progressively finds acceptance in the care of patients undergoing high-risk surgery.[13] The second, thrombin generation (TG), mirrors a significant part of the overall hemostatic system, reflecting contributions of natural procoagulants and anticoagulants to hemostasis and the effect of drugs. Many semiautomated and fully automated assays are on the market today, although regrettably still lacking interlaboratory standardization.[14]

Special attention is required for nonanticoagulant strategies to avoid coagulation of the circuit. Patients with a high hematocrit are at higher risk for clotting of the extracorporeal/dialysis circuit because of the higher viscosity of the blood. Blood products should be administered separate from RRT as much as possible. Prompt reaction to pump alarms is important, avoiding interruption of the blood flow. In this regard, well-functioning vascular access is fundamental. Circuits with a lot of blood-air contact because of the use of drip chambers are especially prone to clotting. There is no evidence pointing towards the efficacy of intermittently rinsing the circuit with saline flushes to prevent clotting.[15]

No Anticoagulation Strategy

Several authors described large series of patients treated without any form of anticoagulation during RRT for AKI (in up to 50%–60%).[8,9] Especially in patients with coagulopathy, an acceptable treatment length can be reached even without anticoagulant.[16] Risking circuit clotting—in the worst case implying the loss of approximately 200 mL

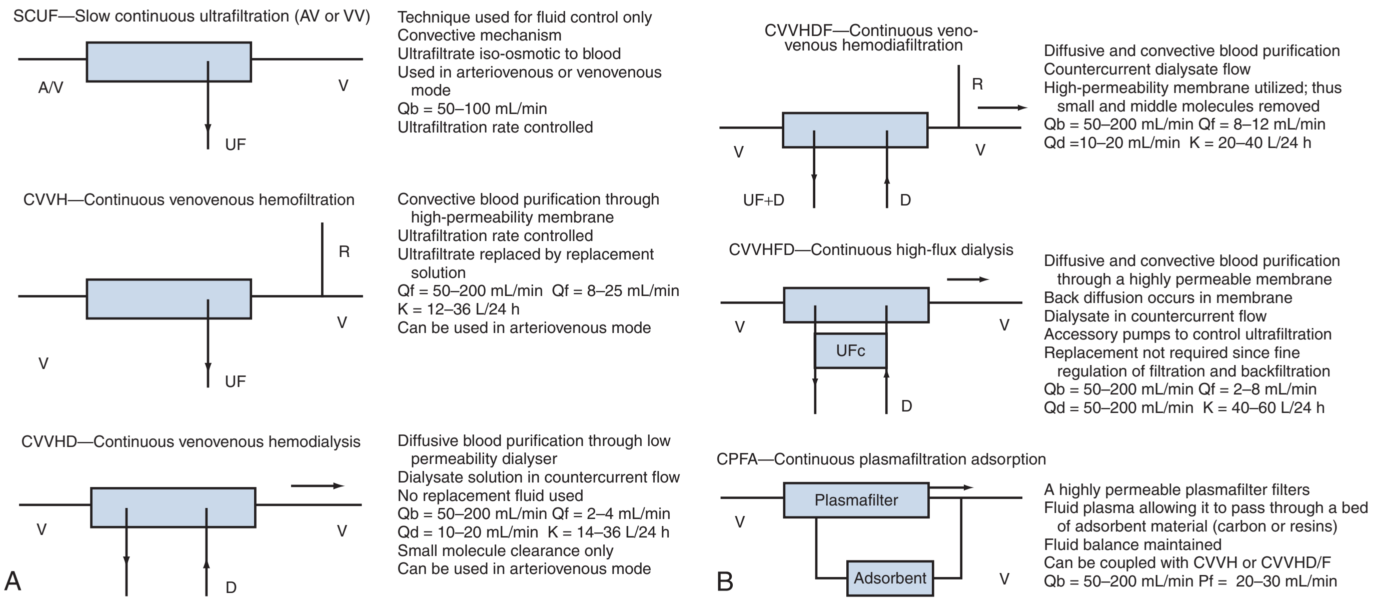

Fig. 38.2 Schematic representation and definitions of the different continuous renal replacement therapies according to standard nomenclature. Functional capabilities are described. *A,* Artery; *D,* dialysate; *K,* clearance; *Pf,* plasma filtration rate; *Qb,* arterial flow; *Qd,* dialysate flow; *Qf,* ultrafiltration rate; *UF,* ultrafiltrate; *UFc,* ultrafiltrate control pump; *V,* vein.

of extracorporeal blood and eventually also the venous access—may be defensible in patients with high bleeding risk. Neither the effect of clogging on filter performance nor the consumption of coagulation factors in RRT without anticoagulation are well studied.

Unfractionated Heparin (UFH)

Current guidelines suggest using UFH for intermittent RRT in patients without increased bleeding risk and in the case of contraindications for citrate in continuous RRT. It has a half-life between 0.5 and 3.0 hours in patients receiving dialysis.[17] It has a rapid action time of approximately 3–5 minutes. UFH acts by potentiating thrombin and inhibiting activated factor Xa. A variety of infusion schemas exists, including single dose, repeated bolus, or continuous infusion. An example of a standard dosing regimen is a bolus of 500–1000 units at the start of treatment, followed by 500–750 units/h. Other authors suggest adapting the dose to the body weight, starting with a bolus of 10–20 units/kg/h, followed by a maintenance infusion of 10–20 units/kg/h that typically would be stopped 30 minutes before the end of treatment. UFH treatment needs to be individualized according to the clinical setting. It can be monitored with routine laboratory tests as activated partial thromboplastin time (aPTT) or activated coagulation time (ACT), but many centers move to anti–factor Xa assays for monitoring UFH because it reflects the anticoagulant activity more adequately, especially during inflammatory conditions.[18,19] Anti–factor Xa level–guided anticoagulation protocols yield important potential in decreasing filter clotting.[20] If routine laboratory tests are used, typical anticoagulation targets in ICU patients are 1.5–2 times prolongation of the aPTT and 40% increase of ACT. If needed, UFH can be reversed with protamine. Heparin failure, resulting in clotting of the circuit, can be the result of antithrombin deficiency or UFH neutralization by binding to plasma proteins. During treatment, the thrombocyte count should be monitored, allowing timely detection of heparin-induced thrombocytopenia (HIT).

Low-Molecular-Weight Heparin (LMWH)

Low-molecular-weight heparins are effective and can safely be used for anticoagulation during RRT in critically ill patients.[21] LMWHs are mostly administered as a single bolus in the beginning of the dialysis, immediately after the start in the case of postfilter administration, or prefilter after 5 minutes.[22] They have a weight-based dosing and are frequently used without therapeutic monitoring in case of short treatment sessions, but especially in the setting of prolonged daily use, periodic measurement of anti–factor Xa levels is mandatory, because LMWHs are only partially cleared during hemodialysis (especially with high-flux membranes).

Citrate

The 2012 Kidney Disease Improving Global Outcomes (KDIGO) Clinical Practice Guidelines for Acute Kidney Injury have recommended regional citrate anticoagulation as the preferred anticoagulation modality for CRRT in critically ill patients for whom it is not contraindicated.[10] Citrate chelates calcium, an essential cofactor of many steps of the coagulation cascade. A whole range of protocols exist, applying various dialysate/replacement fluids for different RRT modalities. Prefilter infusion of citrate, either as a separate trisodium citrate solution or added to a calcium-free predilution replacement fluid, lowers the ionized calcium levels in the extracorporeal circuit to a level that achieves full blood anticoagulation (i.e., 0.3–0.4 mmol/L). In most of the protocols, the citrate dose is adapted according to the result of postfilter circuit ionized calcium measurements, although there is some concern about the reliability of blood gas analyzers at the point of care for this purpose.[23] Having a low molecular weight (198 and 258, respectively) and high sieving coefficient, citrate and citrate complexes are partially removed in the effluent. Citrate clearance is higher with dialysis than with CVVH, with extraction ratios from 20% to 60% depending on the modality and dose of the RRT. In most of the treatment protocols calcium infusion is needed to replace calcium losses and maintain systemic ionized calcium levels within the normal range. Remaining citrate is metabolized in the liver, muscle, and kidney, producing bicarbonate and eventually leading to metabolic alkalosis; therefore most of the commercially available replacement fluids contain a lowered bicarbonate concentration. In patients with impaired citrate metabolism, acidosis can ensue. Other potential metabolic derangements include hypernatremia (metabolization of trisodium citrate), hypomagnesaemia (effluent losses in the form of citrate complexes), and hypocalcemia. Citrate accumulation can occur in patients with a profound shock or severe liver failure, although studies argued that citrate can be used safely in patients with liver dysfunction.[24] In case of citrate accumulation, there is a simultaneous increase of the total calcium and fall of the ionized calcium. The total calcium to ionized calcium ratio is the best marker for citrate intoxication. When this ratio exceeds 2.5, citrate administration must be stopped. Unintended rapid infusion of a hypertonic citrate solution, causing life-threatening hypocalcemia, is the main risk of citrate anticoagulation. In this case, it is recommended to stop the citrate infusion while continuing dialysis with a calcium-containing dialysate. In experienced hands, severe hypocalcemia-related complications seldom occur, and regional citrate anticoagulation has been shown to be safe. Treatment protocols should describe how to adjust flows under different conditions to prevent metabolic derangements. Compared with heparin, citrate is associated with lower risk of circuit loss, lower incidences of filter failure, less bleeding, and lower transfusion rates. Furthermore, citrate is a source of energy and has potential antiinflammatory effects.

UFH and Protamine

Regional anticoagulation can also be accomplished by the combination of UFH and protamine. Its use has decreased in parallel with the increasing popularity of citrate. Protamine has several side effects such as anaphylaxis, hypotension, cardiac depression, leukopenia, and thrombocytopenia. Further, there may be a risk for a rebound anticoagulant effect because of the shorter half-life of protamine compared with heparin. Regional anticoagulation with heparin-protamine is therefore no longer recommended.[10,25,26]

Platelet-Inhibiting Agents

Prostacyclin (PGI_2) and its analogue (nafamostat) inhibit platelet aggregation and adhesion. They have been used alone or in combination with heparin to improve filter survival, but several guidance documents do not recommend their use in RRT. In addition, these drugs are expensive, and there are safety concerns over hemodynamic stability with the use of prostacyclin, and anaphylaxis, agranulocytosis, and hyperkalemia with the use of nafamostat.

Dialysis Access

Central venous access is with a double-lumen catheter, preferably in the right jugular vein or a femoral vein. Use of the subclavian vein is not recommended, because contact of the catheter with the vessel wall may lead to thrombosis and ensuing stenosis, jeopardizing the possibility for an arteriovenous fistula in case there is no recovery of kidney function and the patient remains dialysis dependent. This will more likely occur when the catheter has a trajectory with angulations, such as when inserted in the left jugular vein or the subclavian veins.

For optimal blood flow rates, the tip of the catheter should be located in a large vein (i.e., the inferior or superior vena cava). For an adult, the optimal length of a catheter in the right jugular vein is therefore 12–15 cm; the left jugular vein 15–20 cm; and for the femoral approach, this is 19–24 cm. There are several different designs of dialysis catheters. At present it is not clear which design is preferable. The outer diameters of catheters vary between 11 and 14 French; larger catheter diameters will result in better blood flow rates.[27]

Temporary, noncuffed, nontunneled dialysis catheters are preferred in case of sepsis, lack of image guidance, and uncorrectable coagulopathy. However, in the absence of these contraindications, placement of a cuffed tunneled catheter may be considered immediately or after 2 weeks of using a temporary catheter.[28,29]

Like hemodialysis catheters, the insertion of Tenckhoff catheters for peritoneal dialysis can be done bedside using a Seldinger technique. In challenging cases, ultrasound or fluoroscopic guidance may be of added value. In patients with previous midline surgical scars or suspicion of intraabdominal adhesions, a surgical approach (laparoscopy or laparotomy) that offers direct vision is preferred. In experienced teams, start of peritoneal dialysis within 24–48 hours of catheter placement comes with low rates of PD fluid leaks, and rapid escalation in dwell volumes is possible. Tenckhoff catheters placed using the percutaneous technique or surgically with a purse-string suture can be removed easily, or might serve as a long-term access in case patients fail to recover from the AKI episode.

INITIATION OF RRT

The kidneys are crucial for removal of water and homeostasis of electrolytes and acid-base. Volume overload in an anuric patient and severe electrolyte and acid-base abnormalities are therefore absolute criteria for initiation of RRT (Table 38.2).[30,31] Unlike HD or CRRT, ultrafiltration and clearance cannot be exactly predicted in PD treatment. High-glucose-containing PD solutions can remove up to 1 L of fluid in 4 hours. Standard PD solutions do not contain potassium, but because solute removal is slower during PD, it may take a one day of high-volume PD to achieve serum potassium control, and later on the treatment potassium administration (intravenous or via the PD solution) may be necessary.

The exact timing of initiation of RRT in ICU patients has been the topic of fierce debates for decades. If exposure to RRT was without risks, we would not wait until occurrence of absolute criteria. But very

TABLE 38.2 Criteria for Initiation of RRT for AKI

Indication	Characteristic
ABSOLUTE CRITERIA	
Metabolic abnormality	BUN >100 mg/dL (35.7 mmol/L)
	Hyperkalemia >6 mmol/L with ECG abnormalities
	Hypermagnesemia >8 mEq/L (4 mmol/L) with anuria and absence of deep tendon reflexes
Acidosis	pH <7.15
	Lactic acidosis related to metformin use
Fluid overload	Diuretic resistant
RELATIVE CRITERIA	
Metabolic abnormality	BUN >76 mg/dL (27 mmol/L)
	Hyperkalemia >6 mmol/L
	Dysnatremia
	Hypermagnesemia >8 mEq/L (4 mmol/L)
Acidosis	pH >7.15
Anuria/oliguria	AKI stage 1
	AKI stage 2
	AKI stage 3
Fluid overload	Diuretic sensitive

AKI, Acute kidney injury; *BUN,* blood urea nitrogen; *ECG,* electrocardiogram.
Modified from Gibney N, Hoste E, Burdmann EA, et al. Timing of initiation and discontinuation of renal replacement therapy in AKI: unanswered key questions. *Clin J Am Soc Nephrol.* 2008;3:876–880.

early initiation of RRT in patients will expose these patients to potential hazards associated with insertion of the catheter (blood loss, thrombosis, catheter infection) and exposure to the extracorporeal circuit (air embolism, hypotension, etc.).[32] If RRT offers support in patients with only mild or moderate AKI, early initiation may be beneficial. However, available data suggest no benefit in, for instance, modulation of the inflammatory response,[33,34] and one study even showed harm when RRT was started very early in patients with severe sepsis or septic shock.[35]

Cohort studies on timing of RRT showed a benefit for early initiation of RRT.[36,37] However, the few small prospective randomized studies that evaluated early initiation of RRT could not show a benefit for early initiation of RRT.[31,38,39] On the other hand, cohort studies also showed that late initiation is associated with worse outcomes.[40–42] Recently, four large prospective randomized studies have addressed this in different types of cohorts. Of these, the ELAIN study, a single-center study in predominantly surgical ICU patients, found initiation at AKI KDIGO stage 2 led to reduced 90-day mortality when compared with initiating CRRT at AKI KDIGO stage 3.[43] The three other studies in general ICU and septic ICU patients could not show a mortality difference between early or accelerated initiation at KDIGO stage 2 or 3 compared with standard of care.[44–46] Moreover, they noticed that approximately 40% of patients in the standard group were not treated with RRT. Finally, the largest of the three studies, the STARRT-AKI study that included 3019 patients, found that accelerated initiation was associated with a higher proportion of patients dependent on RRT at 90 days.[46] A meta-analysis in which all prospective studies except STARRT-AKI were included showed that there is no benefit for an accelerated initiation of RRT when there are no urgent indications for RRT, and this strategy may lead to less use of RRT and thus costs.[47]

The decision to initiate RRT is therefore based on clinical criteria. These should incorporate the "demand" for RRT, which includes overload of fluid and solutes, but also severity of acute disease and chronic comorbidities.[48] Knowledge on the deterioration or recovery of kidney function within a certain period would help us in the timing of RRT, especially when there are only relative criteria for initiation of RRT (see Table 38.2). A clinical risk assessment may identify patients at greater risk for further deterioration of kidney function. This may take into account risk factors such as age, chronic kidney disease, and severity of illness of the patient. Other tools that have been explored are measurement of specific kidney biomarkers[49] such as C-C motif chemokine ligand 14 (CCL14)[50] and the furosemide stress test[51]

CHOICE OF RRT MODALITY

Since the introduction of CRRT in the mid-1980s, there has been debate on the optimal modality of RRT. Several relatively small studies and meta-analyses showed that CRRT and IHD are associated with similar patient outcomes.[52,53] However, the number of patients included in individual studies was relatively low, the baseline characteristics of patients were different, and the techniques used (modalities, dose, initiation criteria) varied between studies, making comparisons difficult. Renal outcomes may, however, be better when CRRT is used. Cohort studies and also comparison of the large prospective randomized studies on dose—the ATN and RENAL studies—suggest that CRRT is associated with better renal recovery and less end-stage kidney disease in survivors.[8,9,54–56]

CRRT is performed with a lower extracorporeal blood flow and allows removal of fluid over a longer time and a lower ultrafiltration rate, characteristics that enhance the hemodynamic tolerability of this technique. Hence, CRRT is often recommended for hemodynamically unstable patients,[10] although studies in specialized centers could not demonstrate that IHD was tolerated worse compared with CRRT in shock patients.[57]

Intermittent techniques, however, use fewer resources because they allow for several treatments with one machine per day, and dialysate and replacement fluid is produced by the dialysis machines instead of having to buy these special solutions for CRRT. It will depend on the specific setting on whether the cost of CRRT is greater than that of IHD or vice versa.[58,59] An important argument in favor of IHD and hybrid therapy is that these modalities will allow for mobilization of the patient during the off period. A recent meta-analysis suggests that hybrid therapies are associated with the same outcomes as CRRT, suggesting that increasing the length of intermittent treatment may combine the best of IHD and CRRT.[60] However, these data were compiled on a limited number of patients and most in cohort studies, making this conclusion prone to bias.

SPECIAL INDICATIONS

Heparin-Induced Thrombocytopenia

Up to 3% of heparin-exposed patients develop antibodies directed against the complex of heparin and platelet factor 4, resulting in thrombocytopenia with or without thrombosis. If HIT is likely, all heparin administration, including LMWHs, heparin-coated dialyzer membranes, and catheter locks containing heparin, must be stopped. Guidance documents recommend various options for anticoagulation during RRT in this circumstance: regional citrate anticoagulation or the use of direct thrombin inhibitors (such as argatroban or bivalirudin) or factor Xa inhibitors (such as fondaparinux).[10] Argatroban has a rapid onset of action and an elimination half-life of less than 1 hour. Its clearance is mainly hepatic (it is contraindicated in cases of severe liver failure) and independent of kidney function. Bivalirudin is renally cleared to a small extent (20%) and partially (25%) removed by dialysis.[61] Fondaparinux is excreted exclusively by the kidney[61] and can be removed using high-flux dialysis membranes.[62]

Patients with High Serum Urea Concentration

When applying acute RRT to a highly uremic patient (typically $>$175 mg/dL), precautions should be taken to prevent disequilibrium syndrome. This neurologic condition is characterized by nausea, vomiting, restlessness, and headache and may occasionally progress to seizures, coma, or death. The syndrome is believed to be primarily related to the decreasing osmolarity of the blood after the initiation of RRT, creating an osmotic gradient between the blood and the brain, which is compensated for by an influx of water into the brain compartment. To avoid brain edema caused by large variations in osmolality, several preventive measures can be taken, targeting a reduction in the plasma urea nitrogen of, at most, 40%. The dialysis dose can be reduced by lowering blood flow and dialysate flow, using a small dialyzer, and limiting the length of the treatment. The use of a sodium-enriched dialysate may further reduce the risk.[63]

Patients with Intracranial Hypertension or Cerebral Edema

In patients with acute brain injury, AKI requiring RRT may worsen the neurologic status in several ways. The accumulated urea and solutes diffuse from the blood compartment to the brain cells, thereby increasing water uptake by the brain cells. Dysfunction in the blood-brain barrier reinforces this process. The shift of water into brain tissue as a result of lowered tonicity of plasma with respect to the brain cells (as described in disequilibrium syndrome) may lead to increased intracranial pressure causing cerebral hypoperfusion. This is pronounced by the decreased or absent autoregulation of the cerebral blood flow and eventual hypotension during RRT. Both hypotension and disequilibrium can be avoided by the slow progressive removal of fluids and solutes during CRRT, which in this setting is preferred over intermittent RRT. If the patient is also at increased risk for intracranial bleeding, locoregional citrate anticoagulation is recommended.[26] Gradual solute removal during peritoneal dialysis lowers the potential for disequilibrium syndrome and intracranial fluid shifts, making this a treatment modality to consider in patients with increased intracranial pressure. In case of cerebral edema because of an acute rise of ammonia, prompt RRT is mandatory. In adults, hyperammonemia is defined as a serum concentration above 50 μmol/L or 85 μg/dL, and RRT is generally indicated at ammonia levels exceeding 150 μmol/L or 255 μg/dL. When rapid ammonia clearance is needed, HD or high-dose CRRT are the preferred treatment modalities; PD is less efficient.[64,65]

Patients with Hyponatremia

If patients with severe chronic hyponatremia are treated with conventional RRT, the serum sodium concentration can be expected to increase rapidly, exposing patients to the risk of developing osmotic demyelination. High serum urea concentration may protect the brain against this.[66,67] To avoid osmotic demyelination in patients with chronic hyponatremia, the treatment during a single dialysis session has to be adjusted to provide a rate of correction that does not exceed the generally recommended rate. The easiest way to do this is to choosing low-efficiency RRT such as CVVH and to maintain sodium concentration of the replacement fluid slightly higher than the serum sodium concentration. In the routinely available dialysate/replacement solutions, the variability in sodium content is limited. Adapting the sodium concentration can be established by adding sterile water to the replacement fluid bag. Diluting replacement fluid will also result in decreased potassium and bicarbonate concentrations, and therefore may induce hypokalemia and acidosis. When applying HD treatment, one can make use of the lowest-available sodium setting (130 mmol/L), reduce the blood flow rate markedly (to 2 mL/kg/min), and shorten the dialysis time.[68]

Prevention of Contrast-Induced Nephropathy

A single dialysis session removes 60%–90% of contrast media from the blood,[69,70] and one study argued that periprocedural CRRT may be beneficial in patients with chronic kidney disease.[71] However, meta-analyses including studies in patients without severe chronic kidney disease, could not show a benefit for this strategy.[69,72] Considering the possible complications, cost, logistical difficulties, and uncertain benefit, several guidelines do not recommend RRT for the prevention of contrast-induced nephropathy.[10,73,74]

Patients with Severe Hemodynamic Instability

CVVHD, sustained low-efficiency daily dialysis, or continuous HD[10,26,73–76] seem to be equivalent treatment strategies in terms of mortality, kidney recovery, and fluid removal for hemodynamically unstable patients.[75–78] Treating those patients requires some precautions: less-aggressive ultrafiltration; increasing dialysate sodium and calcium concentrations to 145 mmol/L and 1.5 mmol/L, respectively; adapting the dialysate temperature to obtain isothermal dialysis; connecting afferent and efferent bloodlines simultaneously at the start of the procedure; and raising the blood flow slowly.

Patients with Severe Lactate Acidosis

The key issue in the management of lactic acidosis is to treat the underlying cause. CRRT can be performed in critically ill patients with severe lactic acidosis and AKI.[79] Using continuous hemodialysis with bicarbonate dialysate, lactate concentrations can be lowered and the pH can be corrected. However, no adequately powered randomized clinical trial with clinical outcome endpoints has yet evaluated RRT in this setting.[80] Severe lactate acidosis may preclude PD with lactate-buffered solutions.[81]

CONCLUSION

RRT for AKI is used in up to 10%–15% of ICU patients. Urgent indications include severe hyperkalemia and fluid overload not responsive to diuretic therapy. In the absence of urgent indications, initiation of RRT should be based on clinical judgment, which should include the demand for RRT and the capacity of the kidneys. Early initiation of RRT at AKI stage 2 or 3 does not offer a survival benefit. The modality of RRT chosen will be determined by the availability of modalities, the expertise of the team in handling these, and patient characteristics that may change during the course of illness. Continuous therapies are the most used modality in many ICUs. Intermittent hemodialysis can cause hemodynamic instability and is so preferably used in hemodynamically stable patients. Specialized units also use intermittent hemodialysis and hybrid therapies for hemodynamically unstable patients. PD is used in low- and middle-income countries as a first-choice therapy but can be used also in specific patients in case of a capacity surge, such as during the COVID-19 pandemic. Extracorporeal therapies can be performed without anticoagulation. Anticoagulation strategies most often used are regional citrate anticoagulation, UFH, and LMWHs. For continuous RRT a delivered dose of 20–25 mL/kg/h is recommended. To account for downtime, a prescribed dose of 25–30 mL/kg/h is advised.

KEY POINTS

- Urgent indications for initiation of RRT for AKI include severe hyperkalemia and fluid overload not responsive on diuretic therapy.
  - In the absence of urgent indications, early initiation of RRT does not offer a survival benefit.
  - In the absence of urgent indications, RRT is started on clinical judgment and the balance between the capacity of the kidneys to remove fluid and metabolites and the demand of the patient, which includes comorbidities, severity of acute illness, and retention of fluid and metabolites.
- The choice of the modality of RRT will be determined by the availability of modalities, the familiarity of the team to handle these, and patient characteristics. In hemodynamically unstable patients, continuous therapies will be used more often.
- RRT can be performed without anticoagulation in patients with impaired coagulation or increased bleeding risk. Citrate, UFH, and LMWHs are the most often used anticoagulation strategies.

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The largest prospective randomized study on RRT in ICU patients with AKI at present. This study showed that in patients where clinicians had clinical equipoise for initiation of RRT, accelerated initiation at time of AKI KDIGO stage 2 did not offer a benefit on 90-day survival, and in fact was associated with less recovery of RRT at day 90.

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